

# Librika Platform User Manual

Detailed Functional Specifications & API Call Flows

|                              |                                                           |
|------------------------------|-----------------------------------------------------------|
| <b>Specification Source:</b> | Librika AI Library Blueprint (Slides 3-12)                |
| <b>AI Completions:</b>       | NVIDIA Nemotron completions API (30B parameter reasoning) |
| <b>Security Level:</b>       | Role-Based Access Control (RBAC)                          |
| <b>Document Date:</b>        | June 29, 2026                                             |

# Chapter 1: Detailed Functional Specifications

## 1. High-Speed AI Book Scanner & stock Logic

The scanner subsystem (located at /admin/scanner) allows librarians to capture book cover images and process metadata at high speed. The frontend schedules capture frames, sending base64 image data through asynchronous background threads. With database WAL mode (busy\_timeout=10000ms), the catalog handles parallel inserts. If a book with a matching normalized ISBN is scanned, the database automatically triggers stock increment logic: total\_copies += 1 and available\_copies += 1.

## 2. The Double Fallback Accuracy Engine

To ensure metadata accuracy, the VLM scanner implements a three-tier fallback pipeline:

- **Level 1 (NVIDIA Nemotron VLM):** Extracts title, author, publisher, and ISBN from the base64 image. The backend automatically strips markdown code blocks and reasoning <thought> blocks.
- **Level 2 (Google Books API):** If ISBN or Author is missing, queries Google Books API using the parsed title (q=intitle:[Title]) to enrich fields.
- **Level 3 (DuckDuckGo Crawl + LLM Refinement):** If Google fails (common in regional NCERT/syllabus books), crawls DuckDuckGo search snippets and feeds them to the LLM to refine the entry.

## 3. Subscription Matrix & Interactive Billing

Features a subscription manager that dynamically controls capabilities based on school tiers:

- **FREE Tier:** Limits school to 50 students, 500 catalog books, and disables AI scanner/quizzes.
- **BASIC Tier:** Upgrades to 500 students, 5,000 catalog books, and enables AI features.
- **PROFESSIONAL Tier:** Offers unlimited students, catalog books, and AI features.

Tiers are validated dynamically on every add-user, add-admin, or catalog upload event. The Billing Dashboard (/billing) features a 3D payment simulator (rotates 180 degrees on CVV field focus) and displays a sequential transaction authorization loader (Connecting -> Authorizing -> Finalizing) before committing updates.

## 4. Gamified Reading Point & Leaderboard System

Rewards students for reading and responsibility. Points calculation formula:

$$\text{Reader Score} = (\text{Issues} \times 5) + (\text{Returns on Time} \times 15) + (\text{Approved Reviews} \times 20) + (\text{Passed Quizzes} \times 50) - (\text{Late Returns} \times 20) - (\text{Lost Books} \times 50)$$

The Leaderboard page (/leaderboard) provides three distinct portals: Physical Library, Digital Library, and Overall Aggregated. It supports class and section filters alongside timeframe splits (Weekly, Monthly, Yearly, All-Time). Student Profile Hubs track streaks and award achievement badges (Comprehension Champion, Reading Streak Master, Digital Contributor) based on reading consistency.

## 5. AI Comprehension Quiz System & Anti-Exploit Locks

To verify students have read books before earning leaderboard points, returning a book triggers the quiz engine:

- **Step 1 (Return):** Student returns a completed book.
- **Step 2 (Trigger):** System triggers LLM completions via backend API.
- **Step 3 (Generate):** NVIDIA Nemotron reads book catalog data and generates a 5-question multiple-choice quiz.
- **Step 4 (Evaluate):** Student submits answers. System logs pass/fail statuses, correct counts, and timestamp records.

Awards +50 points on pass.

### Anti-Exploit Security Locks:

- *Minimum Reading Period Lock:* Locks the quiz if a book is returned too quickly (Short books: 2 days; Medium: 5 days; Long: 7 days).
- *90-Day Cooldown:* No duplicate points can be earned for reading the same book within 90 days.

## 6. Community e-Library & Peer Repository

Students upload revision notes, projects, summaries, or articles to /digital-library. A built-in HTML5 reader supports EPUB and PDF files. The system automatically tracks reading progress and syncs reading progress percentage directly to the database.

## 7. Intent-Based AI Search Engine

Integrates a hybrid search engine in catalog views:

- **Local Hybrid Search:** Instant keyword filtering using client-side JavaScript.
- **AI Semantic Search:** Matches queries conceptually using the NVIDIA completions API (e.g. searching 'courage in space' returns astronaut biography and sci-fi books) and renders ■ XX% Match badges.

## 8. The Librika Ecosystem Flywheel

Summarizes the platform cycle: Librarian inputs (Smart AI scanner) enable Student access (3D billing, semantic search, e-Library), which fuels Verified Learning (Comprehension quizzes, progress logs), which drives more reading, completing the loop.

## Chapter 2: Module API Call Flows & Technical Endpoints

### Module A: AI Smart Book Scanner Call Flow

| Step | Call Component / Endpoint      | Action & Output Details                                                                                                                                                          |
|------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | POST /admin/api/upload-cover   | Takes captured camera frame. Resizes image under 600px width (for DB base64 compliance) and saves file. Returns static path URL.                                                 |
| 2    | POST /api/scan-vision          | Passes the cover path. Encodes to base64, queries NVIDIA Nemotron Vision API to extract title, publisher, class, and ISBN. Strips LLM reasoning thought blocks before JSON load. |
| 3    | API Fallback 1: Google Books   | If VLM returns empty critical fields, requests Google Volumes API for catalog data using title query (e.g. isbn:123).                                                            |
| 4    | API Fallback 2: DDG Web Search | If Google fails (NCERT/local books), queries DuckDuckGo for snippets. Passes crawled results to LLM for metadata correction.                                                     |
| 5    | POST /admin/api/save-scanned   | Checks database. If unique ISBN/Title exists, calls /admin/api/add-copy to increment copies. Else, inserts new record and generates Code-128 barcode image.                      |

### Module B: Reader Leaderboard Call Flow

| Step | Call Component / Endpoint      | Action & Output Details                                                                                                   |
|------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1    | Issue Book (Physical/Digital)  | Allocates +5 points to student score and inserts transactions record.                                                     |
| 2    | Return Book (Librarian return) | Evaluates returned date. If before due date: awards +15 points. If late: deducts -20 points. If lost: deducts -50 points. |
| 3    | Submit Review                  | Writes user review record. Once approved by admin moderation: awards +20 points.                                          |
| 4    | AI Quiz Completion             | Checks answers. If passed: awards +50 points to leaderboard totals.                                                       |

## Module C: AI Quiz Comprehension System Call Flow

| Step | Call Component / Endpoint | Action & Output Details                                                                                                                                                                               |
|------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | GET /student/quiz//       | Evaluates quiz eligibility parameters. Locks quiz if minimum reading period is not met based on page length (Short: 2d, Med: 5d, Long: 7d). Checks 90-day cooldown to block duplicate points farming. |
| 2    | ai_generate_quiz()        | If quiz record does not exist in book_quizzes table, sends book metadata to NVIDIA completions API. Formats a 5-question multiple-choice JSON array.                                                  |
| 3    | POST /student/quiz//      | Submits student answers. Grades attempt out of 5. Logs pass/fail metrics. If passed, awards +50 points to student profile and locks future attempts.                                                  |

## Module D: AI Semantic Search Engine Call Flow

| Step | Call Component / Endpoint             | Action & Output Details                                                                                                                               |
|------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | GET /student/browse?q=[query]&ai=true | User checks AI toggle and submits search. Intercepts request in Flask. Queries database for available catalogs.                                       |
| 2    | perform_ai_semantic_search()          | Constructs prompt containing query and book details. Passes parameters to completions endpoint to calculate relevance score (0-100) for each item ID. |
| 3    | Render student_browse.html            | Filters items scoring 0. Sorts list by match percentage descending and renders a gradient ■ XX% Match badge on card overlays.                         |

## Module E: Subscriptions & Billing Portal Call Flow

| Step | Call Component / Endpoint | Action & Output Details                                                                                                                |
|------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Upgrade Select            | Admin opens /billing. System reads limits and lists starter, basic, and professional packages.                                         |
| 2    | Checkout Form Click       | Pops up the 3D payment modal. Intercepts inputs to update card template. Flips 180 degrees on focusing CVV field.                      |
| 3    | POST /billing/checkout    | Simulates transaction authorization, creates invoice logs, sets subscriptionStatus to 'active', and upgrades catalog quotas in SQLite. |